A MULTIFREQUENCY INVERSE SOURCE PROBLEM WITH KERR NONLINEARITY IN INHOMOGENEOUS MEDIA

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Abstract. We study the inverse source problem for the three-dimensional Helmholtz equation with Kerr nonlinearity in an inhomogeneous medium. For sufficiently small sources with certain regularities, we demonstrate that the complex Dirichlet boundary measurements taken on any open frequency interval uniquely determine the source. Moreover, we establish the stability estimate consisting of a Lipschitz discrete data discrepancy at finite eigenfrequencies and a high-frequency tail which decreases as more eigenfrequencies of data are used. In particular, a conditional double-logarithmic estimate is obtained from data collected only in a fixed frequency band. The analysis combines localized resolvent estimates and a two-component holomorphic continuation of the nonlinear Kerr equation. Characterizations of radiating and non-radiating sources are also investigated. For the numerical method, a spectral reconstruction is proposed for the inverse source problem. The convergence analysis for the proposed method is investigated with explicit dependence on noise, spectral truncation, and source-space approximation. Numerical experiments are presented to demonstrate the effectiveness of the proposed method.

1. Introduction

Inverse source scattering problems concern the recovery of sources from observations of the wave fields outside their supports. Such problems arise in antenna design and imaging [2]. In general, an inverse source problem at a fixed frequency is ill-posed: non-radiating sources obstruct uniqueness, and small errors in the measurements may cause large reconstruction errors. Inverse theory for linear wave equations has been well-established. In [1], radiating and non-radiating sources have been characterized through the variational method and an orthogonal decomposition. To overcome the ill-posedness at a single frequency, multi-frequency measurements provide additional information about the source, which regain the uniqueness and stability [5, 6, 13].

Nonlinear wave equations arise naturally in the study of time-harmonic wave propagation in nonlinear media. By assuming a time-periodic dependence of the underlying wave field or source term, a time-dependent nonlinear wave equation can be reduced to a nonlinear Helmholtz-type equation. In nonlinear optics, an important example is the propagation of intense laser beams through Kerr-type dielectric media, in which the refractive index depends on the local intensity of the electric field [7]. Numerical simulations can be found in [3, 4, 16]. Compared with their linear counterparts, inverse source problems for nonlinear wave equations are much less studied. Recently, there are some works on inverse problems for nonlinear partial differential equations (PDEs). However, these works are mainly concerned with recovering the nonlinearities in the equations [8, 10, 11, 12]. Uniqueness results for inverse source problems associated with semilinear PDEs have also been established in the inverse boundary value setting [14, 15], with the Dirichlet-to-Neumann map serving as the measurement data. To the best of

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our knowledge, analogous uniqueness results in the scattering setting of recovering source term remain unavailable. For scattering problems, Dirichlet measurements or far-field patterns are more relevant to practical applications.

In this paper, we consider the Helmholtz equation with Kerr-type nonlinearity in an inhomogeneous medium

$$\Delta u(x,k) + k^2 c^{-2}(x) u(x,k) + k^2 q(x)|u(x,k)|^2 u(x,k) = f(x), \qquad x \in \mathbb{R}^3, \tag{1.1}$$

where $k > 0$ is the frequency, $c(x)$ is the wave speed, and $q(x)$ is the real-valued nonlinear susceptibility. Here $f(x) \in L^2(\mathbb{R}^3)$ is the source function and it is assumed $\mathrm{supp} f \subset \overline{D}$, where D is a smooth bounded open set. We also assume that $c \in C^\infty(\mathbb{R}^3)$, where $0 < c_- \le c(x) \le c_+$ and $c^{-2} - 1 \in C_c^\infty(D)$, and $q \in C_c^\infty(D)$. The radiating field is assumed to satisfy the Sommerfeld radiation condition

$$\lim_{r\to\infty} r(\partial_r u - iku) = 0, \qquad r = |x|, \tag{1.2}$$

uniformly in all directions x/r. Let $B_R = \{x \in \mathbb{R}^3 : |x| < R\}$ such that $\overline{D} \subset B_R$, and denote $\Gamma = \partial B_R$. We are concerned with the inverse problem, which is to determine f from the complex boundary values $g_f(k) = u_f(\cdot, k)|_\Gamma$ given in a finite frequency band.

We first establish well-posedness and derive resolvent estimates for the direct scattering problem under a smallness assumption on the source term. We then prove uniqueness and establish a stability estimate for the inverse source problem. Compared with the linear inverse source problem solved by [13] in an inhomogeneous medium, the main difficulty in the current setting is that the source coefficients are no longer equal to the measured eigenfrequency moments (measurement data). The difference contains the Kerr nonlinear term at a different frequency for each coefficient. To fix this issue, we consider the differences between the spectral source coefficients and the data, and derive a quantitative error estimate. To achieve this, we require the source to have a certain degree of regularity and sufficiently small norms. We also show that the data is holomorphic for frequencies in a strip-domain in the complex plane, and the stability follows by applying an analytic continuation.

We also investigate the inverse source problem at a fixed frequency. It is implied by [1] that for linear operator $L_k = \Delta + k^2 c^{-2}$, the L^2 space can be decomposed into the direct sum of two orthogonal spaces, one contains radiating sources and the other contains non-radiating ones. Hence, given data at a single frequency generated by a source, one can determine the radiating part, while the non-radiating part does not produce wave field outside the observation ball B_R. The proof in [1] uses the variational method, which however is no longer applicable for the current setting due to the nonlinearity of the equation. In this work, we develop a novel method for characterizing radiating and non-radiating sources for nonlinear wave equations with sources in an admissible set.

By studying the stability for the inverse problem, we propose a numerical method for reconstructing the sources for nonlinear equations. We employ the spectral identity in Lemma 3.1 to develop a projected nonlinear reconstruction from Dirichlet data at the background eigenfrequencies. Convergence analysis is also presented, which shows that the developed numerical method converges with an explicit error bound depending on moment noise, spectral truncation, and source-space approximation.

This paper is organized as follows. Sections 2 and 3 establish the direct problem and multifrequency inverse results, respectively, while Section 4 treats single-frequency data. Section 5 develops and tests the reconstruction method. Appendix A contains two auxiliary estimates.

2. Direct scattering problem

In this section, we prove the well-posedness of the direct scattering problem. We also establish the resolvent estimates and investigate continuation of the resolvent to the complex plane.

Let $P = -c^2\Delta$ be the self-adjoint operator on $L^2(\mathbb{R}^3, c^{-2}\,\mathrm{d}x)$ with domain $H^2(\mathbb{R}^3)$. Its principal symbol is $p(x,\xi) = c^2(x)|\xi|^2$. We impose the following geometric condition on the background medium.

Assumption 2.1. *The metric $c^{-2}\,\mathrm{d}x^2$ is nontrapping: every nonzero bicharacteristic of p escapes every bounded spatial set as $t \to \pm\infty$.*

Denote by $R_+(z) = (P - z^2)^{-1}$ the outgoing resolvent, initially defined for $\operatorname{Re} z > 0$ and $\operatorname{Im} z > 0$, and continued meromorphically as an operator from compactly supported functions to local Sobolev spaces. Set

$$\mathcal{S}_+(z)F = -R_+(z)(c^2F), \qquad \mathcal{S}_-(z)F = \overline{\mathcal{S}_+(\overline{z})\overline{F}}. \tag{2.1}$$

For real $k > l$, these operators give the outgoing and incoming solutions, respectively, of $(\Delta + k^2c^{-2})v = F$. Throughout the paper, C denotes a positive constant depending on the fixed coefficients and geometry, but independent of the source and wavenumber (for fixed $l > 0$). Its value may vary from line to line but should be clear from the context.

The following lemma provides the resolvent estimates of the operator P and a strip-type analytic domain.

Lemma 2.2. *There exists $d > 0$ such that $\mathcal{S}_\pm$ are holomorphic in the half-strip*

$$\Omega_d = \{z \in \mathbb{C} : \operatorname{Re} z > l,\ |\operatorname{Im} z| < d\},$$

where $l > 0$ is an arbitrary fixed constant. For F supported in $\overline{D}$, the following estimates hold uniformly in Ω_d:

$$\|\mathcal{S}_\pm(z)F\|_{L^2(B_R)} \le C\langle z\rangle^{-1}\|F\|_{L^2(\mathbb{R}^3)}, \tag{2.2}$$

$$\|\mathcal{S}_\pm(z)F\|_{H^2(B_R)} \le C\langle z\rangle^{-1}\|F\|_{H^2(\mathbb{R}^3)}, \tag{2.3}$$

$$\|\mathcal{S}_\pm(z)F\|_{H^2(B_R)} \le C\langle z\rangle^{-2}\|F\|_{H^4(\mathbb{R}^3)}, \tag{2.4}$$

where $\langle z\rangle = (1 + |z|^2)^{1/2}$.

Proof. Since the outgoing resolvent $\chi R_+(z)\chi$ has no poles at positive real wavenumbers, from the results in [9, Theorem 4.43] we have that for each $l > 0$, if d is sufficiently small, the resolvent $\chi R_+(z)\chi$ is holomorphic in Ω_d with the resolvent estimates

$$\|\chi R_+(z)\chi\|_{L^2\to L^2} \le C|z|^{-1}e^{T(\operatorname{Im} z)_-}.$$

Here χ is a smooth compactly supported cutoff function equal to one on a neighborhood of B_R, and $(t)_- = \max\{0, -t\}$. Thus, we prove (2.2).

For $G \in H^2(\mathbb{R}^3)$ supported in $\overline{D}$, we have the identity $PR_+(z)G = R_+(z)PG$. Letting $B_{R'}$ be a ball with $R' > R$, by using the interior elliptic regularity theory we have the estimate

$$\begin{aligned}\|R_+(z)G\|_{H^2(B_R)} &\le C\Big(\|R_+(z)PG\|_{L^2(B_{R'})} + \|R_+(z)G\|_{L^2(B_{R'})}\Big)\\ &\le C\langle z\rangle^{-1}\|G\|_{H^2(\mathbb{R}^3)}.\end{aligned}$$

Hence by taking $G = c^2F$ we prove (2.3). For $F \in H^4(\mathbb{R}^3)$, the resolvent identity further yields

$$\mathcal{S}_+(z)F = z^{-2}c^2F - z^{-2}R_+(z)P(c^2F). \tag{2.5}$$

Using estimates (2.2) and (2.3), the $H^2(B_R)$ norms of the two terms on the right hand side of (2.5) can be bounded by $C|z|^{-2}\|F\|_{H^2}$ and $C|z|^{-2}\langle z\rangle^{-1}\|F\|_{H^4}$, respectively, which yields (2.4). The same results also hold for $\mathcal{S}_-$ by using the conjugation identity (2.1). The proof is complete. □

The Helmholtz equation considered in this paper has a Kerr-type nonlinearity. To continue the solution at positive frequencies to the complex ones, we introduce two independent unknowns and consider the system

$$\begin{cases} U = \mathcal{S}_+(z)[f - z^2 q U^2 V], \\ V = \mathcal{S}_-(z)[\overline{f} - z^2 q V^2 U]. \end{cases} \tag{2.6}$$

To investigate the system, we assume that f belongs to the following a priori function space:

$$\mathcal{A}_\rho = \{f \in H^4(\mathbb{R}^3;\mathbb{C}) : \operatorname{supp} f \subset \overline{D},\ \|f\|_{H^4(\mathbb{R}^3)} \le \rho\},$$

where ρ is a sufficiently small constant. We have the following results.

Theorem 2.3. *There exist constants $A > 0$ and $\rho_* > 0$ such that, for $f \in \mathcal{A}_\rho$ and $0 < \rho \le \rho_*$, the system (2.6) has a unique solution in the product of the two closed $H^2(B_R)$ balls of radius $A\rho\langle z\rangle^{-2}$ in $H^2(B_R)\times H^2(B_R)$ for $z \in \Omega_d$. This solution is holomorphic in Ω_d and satisfies the estimate*

$$\|U(z)\|_{H^2(B_R)} + \|V(z)\|_{H^2(B_R)} \le C\rho\langle z\rangle^{-2}. \tag{2.7}$$

Moreover, for real $k > l$, one has $V(k) = \overline{U(k)}$, and $U(k)$ extends to a solution of the scattering problem (1.1)–(1.2).

Proof. Notice that the space $H^2(B_R)$ is a Banach algebra and is continuously embedded in $L^\infty(B_R)$. Since q is compactly supported in D, the zero extension of qU^2V belongs to $H^2(\mathbb{R}^3)$ and is bounded by

$$\left\|qU^2V\right\|_{H^2(\mathbb{R}^3)} \le C\|U\|^2_{H^2(B_R)}\|V\|_{H^2(B_R)}. \tag{2.8}$$

The existence of the solution to the system is proved by fixed point theory in $H^2(B_R)\times H^2(B_R)$. To do this, we consider the product $\mathcal{B}\times\mathcal{B}$, where $\mathcal{B} := \{U \in H^2(B_R) : \|U\|_{H^2(B_R)} \le r\}$ denoting the ball of radius r in $H^2(B_R)$. By direct calculations in view of (2.8), the associated difference $(U,V) - (\widetilde{U},\widetilde{V})$ has Lipschitz constant Cr^2, with the maximum norm on the product space. Moreover, by (2.3)–(2.4), the linear terms in (2.6) are bounded by $C\rho\langle z\rangle^{-2}$. For $r = A\rho\langle z\rangle^{-2}$, the nonlinear Lipschitz constant is at most

$$C\langle z\rangle^{-1}|z|^2r^2 \le CA^2\rho^2\langle z\rangle^{-3}. \tag{2.9}$$

Then we choose A such that the linear terms have norm at most $r/2$, and choose ρ_* so that (2.9) is at most $1/2$. As a consequence, the right-hand side of (2.6) preserves the product of balls and is a contraction. Using the fixed point theory, this proves existence, uniqueness in these balls, and (2.7).

Regarding the analyticity, the iterates starting from $(0,0)$ are holomorphic and converge locally uniformly in Ω_d. Hence their limit is also holomorphic. For real k, since q is real-valued, by taking complex conjugation of the system (2.6) and repeating the above arguments, we have $V(k) = \overline{U(k)}$ by uniqueness of the system. Applying $\mathcal{S}_+(k)$ to the compactly supported effective source $f - k^2q|U|^2U$ in the first equation provides the required outgoing extension. □

Remark 2.4. The H^4 regularity assumption on the source yields the k^{-2} decay in (2.7). This ensures that the nonlinear spectral remainder obtained below is square-summable over the three-dimensional eigenfrequency sequence.

As a consequence, the direct problem is well-posed on the small outgoing branch. The uniqueness below is restricted to the indicated small ball.

Theorem 2.5. *Fix $l > 0$. For $f \in \mathcal{A}_\rho$ and $0 < \rho \le \rho_*$, the scattering problem (1.1)–(1.2) has an outgoing solution $u_f(\cdot, k) \in H^2_{\mathrm{loc}}(\mathbb{R}^3)$ for every $k > l$. It is unique among outgoing solutions satisfying $\|u(\cdot, k)\|_{H^2(B_R)} \le A\rho\langle k\rangle^{-2}$, and*

$$\|u_f(k)\|_{H^2(B_R)} \le C\rho\langle k\rangle^{-2}. \tag{2.10}$$

For $f_1, f_2 \in \mathcal{A}_\rho$, these solutions satisfy

$$\|u_{f_1}(k) - u_{f_2}(k)\|_{H^2(B_R)} \le C\langle k\rangle^{-2}\|f_1 - f_2\|_{H^4(\mathbb{R}^3)}, \qquad k > l. \tag{2.11}$$

The constants are independent of k and the sources.

Proof. Existence and (2.10) follow from Theorem 2.3. Every outgoing solution in the indicated ball satisfies the first resolvent equation in (2.6). Its conjugate satisfies the second. Uniqueness of the fixed point therefore proves uniqueness in this ball.

Let (U_i, V_i) be the pairs associated with f_i, and put

$$\delta(z) = \max\{\|U_1(z) - U_2(z)\|_{H^2(B_R)}, \|V_1(z) - V_2(z)\|_{H^2(B_R)}\}.$$

Subtracting the two fixed point equations and using (2.4) and the contraction bound (2.9) gives

$$\delta(z) \le C\langle z\rangle^{-2}\|f_1 - f_2\|_{H^4(\mathbb{R}^3)} + \tfrac{1}{2}\delta(z).$$

Absorbing the last term and taking real $z = k > l$ proves (2.11). □

3. Inverse scattering problem

In this section, we consider the inverse source problem. We begin by relating the boundary data to the spectral coefficients of the source. To this end, let P_D be the Dirichlet realization of $-c^2\Delta$ in the weighted L^2-space $\mathcal{H} = L^2(B_R, c^{-2}\,dx)$, and let $\{\kappa_j^2\}_{j=1}^\infty$ and $\{\phi_j\}_{j=1}^\infty$ be the corresponding eigenvalues and eigenfunctions such that

$$P_D\phi_j = \kappa_j^2\phi_j, \qquad \phi_j|_\Gamma = 0, \qquad (\phi_j, \phi_\ell)_{\mathcal{H}} = \delta_{j\ell}. \tag{3.1}$$

The eigenfrequencies are ordered increasingly with multiplicity. For the multifrequency analysis, fix $0 < l < \min\{a, \kappa_1\}$, where $I = (a, b) \subset (0, \infty)$ is the measured band. All fields below belong to the small outgoing branch of Theorem 2.5. Thus, for $h \in L^2(B_R)$, denoting $h_j = \int_{B_R} h\overline{\phi_j}\,dx$ we introduce the norm in the weighted L^2-space

$$\|h\|_c^2 = \int_{B_R} c^2|h|^2\,dx = \sum_{j\ge1} |h_j|^2.$$

We also introduce the function space $\mathcal{X}$ containing functions in $L^2(B_R)$ with norm

$$\|h\|_{\mathcal{X}}^2 = \sum_{j\ge1}\langle\kappa_j\rangle^8|h_j|^2 < \infty, \quad h \in L^2(B_R), \quad \langle\kappa_j\rangle = (1 + \kappa_j^2)^{1/2},$$

which is indeed $\|h\|_{\mathcal{X}} = \|(I + P_D)^2(c^2h)\|_{\mathcal{H}}$, and the weighted norm $\|h\|_c$ is equivalent to the standard one $\|h\|_{L^2(B_R)}$.

Let $f_1, f_2 \in \mathcal{A}_\rho$ be two sources, and let $u_{f_1}(k)$ and $u_{f_2}(k)$ denote the corresponding radiating solutions. Denote the differences by $h = f_1 - f_2$ and $w(k) = u_{f_1}(k) - u_{f_2}(k)$. Further, we denote

$$b_j = -\int_\Gamma w(x, \kappa_j)\partial_\nu\overline{\phi_j}(x)\,dS, \qquad D_N^2 = \sum_{j=1}^N |b_j|^2,$$

and the discrete data discrepancy by

$$E_N^2 = \sum_{j=1}^{N} \kappa_j^2 \|w(\kappa_j)\|_{L^2(\Gamma)}^2.$$

We use the conventions $D_0 = E_0 = 0$ and $\kappa_0 = 0$.

The following lemma deals with estimating the difference between the j-th spectral coefficient h_j of the source and the boundary data b_j.

Lemma 3.1. *Let $e_j = h_j - b_j$. There exists $\rho_* > 0$, independent of N, such that, for $0 < \rho \le \rho_*$ the following estimate holds*

$$\|\{e_j\}_{j\ge1}\|_{\ell^2} \le \theta \|h\|_c, \tag{3.2}$$

where $\|\{e_j\}_{j\ge1}\|_{\ell^2} = (\sum_{j=1}^{\infty} |e_j|^2)^{1/2}$, and

$$\theta = C_* \rho^2 \left(\sum_{j\ge1} \langle \kappa_j \rangle^{-6} \right)^{1/2} \le \tfrac{1}{2}.$$

Moreover, one has $D_N \le C E_N$ and

$$(1-\theta)\|h\|_c \le D_\infty \le (1+\theta)\|h\|_c. \tag{3.3}$$

Proof. For real $k > l$, write

$$n(k) = k^2 q\Big(|u_{f_1}(k)|^2 u_{f_1}(k) - |u_{f_2}(k)|^2 u_{f_2}(k) \Big).$$

Then we have $w(k) = \mathcal{S}_+(k)[h - n(k)]$. The embedding $H^2(B_R) \hookrightarrow L^\infty(B_R)$ and Theorem 2.3 imply

$$\|n(k)\|_{L^2(B_R)} \le C\rho^2 k^2 \langle k \rangle^{-4} \|w(k)\|_{L^2(B_R)} \le C\rho^2 \langle k \rangle^{-2} \|w(k)\|_{L^2(B_R)}.$$

By applying the L^2 estimate of Lemma 2.2 and decreasing ρ_* the term containing w can be absorbed, which yields

$$\|w(k)\|_{L^2(B_R)} \le C\langle k \rangle^{-1} \|h\|_c, \qquad \|n(k)\|_{L^2(B_R)} \le C\rho^2 \langle k \rangle^{-3} \|h\|_c. \tag{3.4}$$

Then from Green's identity at $k = \kappa_j$ we obtain

$$\int_{B_R} (h - n(\kappa_j)) \overline{\phi_j}\, dx = -\int_\Gamma w(\kappa_j) \partial_\nu \overline{\phi_j}\, dS.$$

Hence, we have $h_j = b_j + e_j$, where $e_j = \int_{B_R} n(\kappa_j)\overline{\phi_j}\, dx$. Since $\|\phi_j\|_{L^2(B_R)} \le c_+$, in light of (3.4) we obtain

$$|e_j| \le C_* \rho^2 \langle \kappa_j \rangle^{-3} \|h\|_c.$$

From Lemma A.2 one has $\|\partial_\nu \phi_j\|_{L^2(\Gamma)} \le C\kappa_j$ and $\kappa_j \asymp j^{1/3}$, and thus the sum $\sum_j \langle \kappa_j \rangle^{-6}$ is finite, which proves (3.2) after another reduction of ρ_*. We also have $|b_j| \le C\kappa_j \|w(\kappa_j)\|_{L^2(\Gamma)}$ and thus $D_N \le C E_N$. Finally, $\{b_j\}_{j\ge1} = \{h_j\}_{j\ge1} - \{e_j\}_{j\ge1} \in \ell^2$, and using the triangle inequalities we prove (3.3). □

Now we introduce continuous data discrepancy

$$\epsilon_I = \sup_{k\in I} k \|w(k)\|_{L^2(\Gamma)},$$

Here $I = (a, b)$ is the frequency band such that $0 < l < a < b < \infty$. Denote

$$M = C_0 \rho, \quad \eta = \epsilon_I / M,$$

where C_0 is a fixed sufficiently large constant such that $0 \le \eta \le 1$.

The following lemma concerns the analytic continuation of the given data to that at large frequencies. The proof utilizes the analytic continuation principle given by

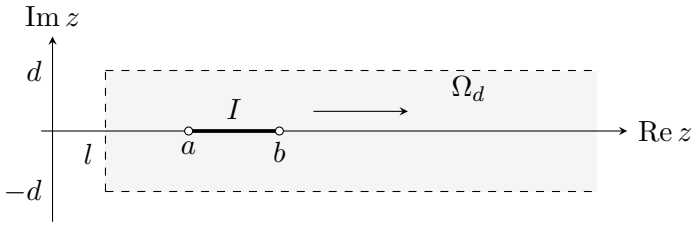


FIGURE 1. The half-strip Ω_d and the measured interval I. Analytic continuation propagates the boundary data along the real axis.

Lemma A.1, and we illustrate the analytic continuation domain in Figure 1. Notice that the estimate (A.1) also holds for eigenfrequencies below the observed band (a, b).

Lemma 3.2. *Let $K > 0$ be an arbitrary constant with $K > \kappa_1$. There exist constants A_0 and $B_0 > 0$ such that for $0 < \eta < 1$ and $\kappa_1 \le k \le K$ one has*

$$k\|w(k)\|_{L^2(\Gamma)} \le M\eta^{A_0 e^{-B_0 K}}. \tag{3.5}$$

The constants depend only on the fixed complex strip, I and κ_1.

Proof. Let $U_i(z)$ denote the outgoing component of the holomorphic pair associated with f_i. By Theorem 2.3 and continuity of the trace operator, the $L^2(\Gamma)$-valued function

$$H(z) = z\Big(U_1(z) - U_2(z)\Big)|_\Gamma$$

is holomorphic in Ω_d. The trace theorem gives

$$\sup_{z\in\Omega_d} \|H(z)\|_{L^2(\Gamma)} \le C \sup_{z\in\Omega_d} |z|\|U_1(z) - U_2(z)\|_{H^2(B_R)} \le C\rho \sup_{z\in\Omega_d} |z|\langle z\rangle^{-2} \le M,$$

where C_0 in $M = C_0\rho$ is fixed sufficiently large. On the real measurement interval, $\sup_{k\in I} \|H(k)\|_{L^2(\Gamma)} = \epsilon_I$. Lemma A.1, with $Y = L^2(\Gamma)$, proves (3.5). □

Combining the analyticity of the data and Lemma 3.1, we can now prove the uniqueness of the inverse source problem from data collected in an arbitrary open frequency interval.

Theorem 3.3. *Let $0 < \rho \le \rho_*$ and $f_1, f_2 \in \mathcal{A}_\rho$. If $u_{f_1}(k)|_\Gamma = u_{f_2}(k)|_\Gamma$ for every $k \in I$, then $f_1 = f_2$.*

Proof. The holomorphic trace difference vanishes on I. By the identity theorem it vanishes throughout the connected domain Ω_d. Thus we have $b_j = 0$ for all $j \ge 1$. Equation (3.3) implies $(1 - \theta)\|f_1 - f_2\|_c = 0$, which gives $f_1 = f_2$. □

The following spectral tail estimate relies on the regularity of the source and will be used in the stability analysis. The proof follows directly by inserting the weights $\langle\kappa_j\rangle^8$ in the sums and using the a priori bound $\|h\|_{\mathcal{X}} \le 2Q$. For the class $\mathcal{A}_\rho$, one may take $Q = C\rho$.

Lemma 3.4. *Assume $\|f_i\|_{\mathcal{X}} \le Q$, $i = 1, 2$. For every integer $N \ge 0$ and real $K \ge 0$,*

$$\sum_{j>N} |h_j|^2 \le 4Q^2\langle\kappa_{N+1}\rangle^{-8}, \qquad \sum_{\kappa_j>K} |h_j|^2 \le 4Q^2\langle K\rangle^{-8}. \tag{3.6}$$

We have the following increasing stability estimates for the inverse source problem.

Theorem 3.5. *Let $0 < \rho \le \rho_*$, $f_1, f_2 \in \mathcal{A}_\rho$, and $\|f_i\|_{\mathcal{X}} \le Q$. We have the following estimates for every $N \ge 0$:*

$$\|h\|_c \le \frac{D_N + 2Q\langle\kappa_{N+1}\rangle^{-4}}{1 - \theta}. \tag{3.7}$$

Moreover, for sufficiently small $\eta > 0$, we have

$$\|h\|_c^2 \le \frac{C}{(1-\theta)^2}\left(E_N^2 + \frac{Q^2+M^2}{\left(\max\{\langle\kappa_N\rangle, \log|\log\eta|\}\right)^8}\right). \tag{3.8}$$

The constants are independent of N, K, and the data discrepancy.

Proof. Since $h_j = b_j + e_j$, we have $|h_j| \le |b_j| + |e_j|$. Then, using (3.2) and Lemma 3.4, we obtain

$$\|h\|_c \le \Big(\sum_{j=1}^{N}|h_j|^2 + \sum_{j\ge N+1}^{\infty}|h_j|^2\Big)^{1/2} \le D_N + \theta\|h\|_c + 2Q\langle\kappa_{N+1}\rangle^{-4},$$

which proves (3.7).

Let $J(K) = \#\{j : \kappa_j \le K\}$ be the counting function of multiplicities. Choose $K \ge \max\{1, \kappa_N\}$. Then $J(K) \le C(1+K)^3$ and $J(K) \ge N$. For $N < j \le J(K)$, Lemma 3.2 and the boundary normal derivative estimate yield

$$|b_j| \le C\kappa_j\|w(\kappa_j)\|_{L^2(\Gamma)} \le CM\eta^{A_0e^{-B_0K}}.$$

Split $\{h_j\}_{j=1}^{\infty}$ into the first N coordinates $\{h_j\}_{j=1}^{N}$, the medium coordinates $\{h_j\}_{j=N+1}^{J(K)}$, and the tail $\{h_j\}_{j=J(K)+1}^{\infty}$. By applying (3.2) to the first two parts and (3.6) to the last part, we have

$$(1-\theta)\|h\|_c \le CE_N + CM(1+K)^{3/2}\eta^{A_0e^{-B_0K}} + 2Q\langle K\rangle^{-4}.$$

By squaring the above estimate, we obtain

$$\|h\|_c^2 \le \frac{C}{(1-\theta)^2}\left(E_N^2 + M^2(1+K)^3\eta^{2A_0e^{-B_0K}} + Q^2\langle K\rangle^{-8}\right). \tag{3.9}$$

To prove (3.8), put $L = |\log\eta|$ and $T = (2B_0)^{-1}\log L$. Let η be sufficiently small and $T \ge 1$. If $\kappa_N \le T$, we choose $K = T$ in (3.9). Then by $\eta^{2A_0e^{-B_0T}} = e^{-2A_0\sqrt{L}}$ the second term in the right-hand side of (3.9) can be bounded by

$$(1+T)^3e^{-2A_0\sqrt{L}} \le C\langle T\rangle^{-8}.$$

Moreover, as the last term is bounded by $C(Q^2+M^2)(\log L)^{-8}$, we have (3.8) in this case. If $\kappa_N > T$, using (3.7) and $D_N \le CE_N$ we have

$$\|h\|_c^2 \le \frac{C}{(1-\theta)^2}\left(E_N^2 + Q^2\langle\kappa_N\rangle^{-8}\right).$$

Moreover, as T is a fixed positive multiple of $\log L$, we prove (3.8). Combining the cases the proof is complete. □

Remark 3.6. For discrete eigenfrequency data, (3.7) separates the measured moment discrepancy from the unresolved spectral tail. Increasing the largest measured eigenfrequency decreases the tail, whereas the accumulated data discrepancy may increase. Thus a decreasing total error bound requires control of the data error as the frequency range is enlarged. This estimate does not establish uniqueness from a finite set of measurements on the full admissible source class.

Remark 3.7. When only the fixed-band data are available, set $N = 0$ in (3.8). This yields a conditional double-logarithmic estimate as $\epsilon_I \to 0$, and exact fixed-band data determine the source uniquely by the preceding uniqueness theorem. When additional discrete eigenfrequency data are available, the maximum in (3.8) combines the measured frequency cutoff with the cutoff obtained by analytic continuation. The discrete data error must still be controlled as this cutoff increases.

4. Radiating and non-radiating sources

In this section, we investigate the inverse source problem at a fixed frequency. We provide an explicit characterization of the non-radiating source, and show uniqueness of the inverse problem for sources with small norms. Compared with their linear counterparts, it can be seen that characterizations of the sources for nonlinear wave equations are much more complicated. For example, we no longer have the elegant orthogonal decomposition of the L^2 source as the linear case, which consists of radiating and non-radiating sources, respectively, as the variational method is not applicable to the nonlinear cases.

Fix $k > l$ and write $L_k = \Delta + k^2 c^{-2}$. A source is called non-radiating at k if the generated radiating field vanishes outside B_R, or equivalently if $g_f(k) = 0$. Otherwise it is called radiating. For $F \in L^2(D)$ we define

$$T_k F = \operatorname{Tr}_\Gamma \mathcal{S}_+(k)F,$$
$$\mathcal{H}_k = \overline{\{v|_D : v \in H^2(B_R),\ L_k v = 0 \text{ in } B_R\}}^{L^2(D)},$$

where the notation $\overline{\{\cdot\}}^{L^2(D)}$ denotes taking closure under the L^2 norm. Let Π_k be the orthogonal projection onto $\mathcal{H}_k$ in the unweighted space $L^2(D)$.

For the linear Helmholtz equation $L_k u = F$, the following lemma identifies the non-radiating subspace and the radiating component in the orthogonal source decomposition. Related decompositions for Maxwell equations were studied in [1]. We give the scalar proof for the present model.

Lemma 4.1. *One has* $\ker T_k = \mathcal{H}_k^\perp$*, and this kernel is infinite-dimensional. Consequently,* $T_k F$ *uniquely determines* $\Pi_k F$*, which is the unique element of minimum* $L^2(D)$ *norm among all* $G \in L^2(D)$ *such that* $T_k G = T_k F$*.*

Proof. Extend F by zero and set $u = \mathcal{S}_+(k)F$. If $T_k F = 0$, uniqueness of the exterior Dirichlet radiation problem implies $u = 0$ outside B_R. Both Cauchy traces on Γ vanish, so $u|_{B_R} \in H_0^2(B_R)$. For every $v \in H^2(B_R)$ satisfying $L_k v = 0$, Green's identity yields

$$\int_D F\overline{v}\,\mathrm{d}x = 0.$$

Continuity extends this orthogonality to the closure defining $\mathcal{H}_k$, and hence $\ker T_k \subset \mathcal{H}_k^\perp$.

Conversely, let $G_k(x, y)$ be the outgoing Green kernel of L_k, normalized by $L_{k,x} G_k(x, y) = \delta_y$. For real coefficients, Green's identity and the radiation condition give reciprocity $G_k(x, y) = G_k(y, x)$. For fixed $|x| > R$, the function $y \mapsto \overline{G_k(x, y)}$ is an $H^2(B_R)$ solution of the homogeneous equation, since its singularity lies outside B_R. If $F \in \mathcal{H}_k^\perp$, the Green representation therefore gives

$$u(x) = \int_D G_k(x, y)F(y)\,\mathrm{d}y = 0, \qquad |x| > R.$$

Thus $T_k F = 0$, proving the reverse inclusion.

The kernel contains $\{L_k\psi : \psi \in C_c^\infty(D)\}$. The map $\psi \mapsto L_k\psi$ is injective on this space, since a compactly supported homogeneous solution is an outgoing solution and must vanish. The kernel is therefore infinite-dimensional. Finally, the sources with the same datum as F form the affine space $\Pi_k F + \mathcal{H}_k^\perp$. Orthogonality gives the unique minimum norm representative $\Pi_k F$. □

For the boundary value $g = T_k F$, let Λ_k^+ denote the Dirichlet-to-Neumann map such that $\Lambda_k^+ g = (\partial_\nu \mathcal{S}_+(k)F)|_\Gamma$. Using Green's identity and the above lemma we have

$$\int_D \Pi_k F\,\overline{v}\,\mathrm{d}x = \int_\Gamma \big((\Lambda_k^+ g)\overline{v} - g\partial_\nu\overline{v}\big)\,\mathrm{d}S, \qquad v \in H^2(B_R), \quad L_k v = 0. \tag{4.1}$$

The volume pairing extends to $\mathcal{H}_k$ by continuity.

To investigate the nonlinear counterpart of the inverse source problem in this paper, we introduce the definition of effective source, which is $F_k(f) = f - k^2 q|u_f(k)|^2 u_f(k)$. Based on the results on linear equations, the following lemma provides the decomposition of the effective source.

Lemma 4.2. *Let $0 < \rho \le \rho_*$ and $f \in \mathcal{A}_\rho$. The effective source has the unique orthogonal decomposition*

$$F_k(f) = f - k^2 q|u_f(k)|^2 u_f(k) = F_k^{\mathrm{r}}(f) + F_k^{\mathrm{n}}(f),$$
$$F_k^{\mathrm{r}}(f) = \Pi_k F_k(f) \in \mathcal{H}_k, \qquad F_k^{\mathrm{n}}(f) = (I - \Pi_k) F_k(f) \in \mathcal{H}_k^{\perp}.$$

The datum $g_f(k)$ uniquely determines $F_k^{\mathrm{r}}(f)$ through (4.1). $F_k^{\mathrm{n}}(f)$ produces no boundary wave field. Moreover, one has

$$g_{f_1}(k) = g_{f_2}(k) \quad \Longleftrightarrow \quad F_k^{\mathrm{r}}(f_1) = F_k^{\mathrm{r}}(f_2).$$

In particular, f is nonradiating if and only if $F_k(f) \in \mathcal{H}_k^{\perp}$. The component $F_k^{\mathrm{r}}(f)$ is the unique minimum-$L^2(D)$-norm effective source producing $g_f(k)$.

Proof. The nonlinear Helmholtz equation gives $u_f(k) = \mathcal{S}_+(k) F_k(f)$, so that $g_f(k) = T_k F_k(f)$. Then the proof follows by using Lemma 4.1 and (4.1). □

In what follows, we first provide an explicit characterization of the non-radiating sources in the space X, where

$$X = \{F \in H^4(\mathbb{R}^3; \mathbb{C}) : \operatorname{supp} F \subset \overline{D}\}.$$

We identify its elements with their restrictions in $L^2(D)$. Notice that in the study of the stability, we use this Banach space for the source functions.

Lemma 4.3. *Fix $k > l$ and $0 < \rho \le \rho_*$. There are neighborhoods $\mathcal{U}_k, \mathcal{W}_k$ of zero in X, with $\mathcal{W}_k \subset \mathcal{A}_\rho$, on which*

$$\Phi_k(F) = F + k^2 q|\mathcal{S}_+(k)F|^2 \mathcal{S}_+(k)F \tag{4.2}$$

is a real analytic diffeomorphism from $\mathcal{U}_k$ onto $\mathcal{W}_k$. Its inverse is $\Theta_k(f) = F_k(f)$, and $g_{\Phi_k(F)}(k) = T_k F$. For each $F_0 \in \mathcal{U}_k$, the entire data fiber in this neighborhood is

$$\{f \in \mathcal{W}_k : g_f(k) = g_{\Phi_k(F_0)}(k)\} = \Phi_k\Big(\mathcal{U}_k \cap (F_0 + \mathcal{H}_k^{\perp})\Big).$$

In particular, the nonradiating sources in $\mathcal{W}_k$ are exactly $\Phi_k(\mathcal{U}_k \cap \mathcal{H}_k^{\perp})$.

Proof. Using local elliptic regularity we have that the mapping $\mathcal{S}_+(k) : X \to H^6_{\mathrm{loc}}(\mathbb{R}^3)$ is bounded. Since $q \in C_c^\infty(D)$, the nonlinear term in (4.2) is a continuous real cubic polynomial from X to itself. Thus Φ_k is real analytic and $D\Phi_k(0) = I$. The real analytic inverse function theorem provides the local inverse.

Write $f = \Phi_k(F)$ and $v = \mathcal{S}_+(k)F$. Then v is an outgoing solution of the Kerr equation with source f. Shrink the neighborhoods so that $\|f\|_{H^4} \le \rho$ and $\|\Phi_k(F) - F\|_{H^4} \le \|F\|_{H^4}/3$. The choice of A in Theorem 2.3 yields

$$\|v\|_{H^2(B_R)} \le \frac{A}{2}\langle k\rangle^{-2} \|F\|_{H^4} \le \frac{3A}{4}\langle k\rangle^{-2} \|f\|_{H^4}.$$

Thus $(v, \overline{v})$ belongs to the prescribed branch ball, and uniqueness of the solution gives $v = u_f(k)$. Consequently, $\Theta_k(f) = F_k(f)$ and $g_{\Phi_k(F)}(k) = T_k F$. The fiber formula now follows from $\ker T_k = \mathcal{H}_k^{\perp}$. □

As a corollary, we have the following theorem, which provides the characterization of the non-radiating sources in the setting of Lemma 4.3.

Theorem 4.4. *Let $f \in \mathcal{A}_\rho$, $0 < \rho \le \rho_*$, and $k > l$. Then f is nonradiating at k if and only if there exists $v \in H_0^2(B_R)$, extended by zero to $\mathbb{R}^3$, such that*

$$f = L_k v + k^2 q|v|^2 v, \qquad \|v\|_{H^2(B_R)} \le A\rho\langle k\rangle^{-2}. \tag{4.3}$$

Proof. If $g_f(k) = 0$, exterior Dirichlet uniqueness gives $u_f(k) = 0$ outside B_R. Its two Cauchy traces vanish, so its restriction belongs to $H_0^2(B_R)$ and satisfies (4.3). Conversely, the zero extension of v lies in $H^2(\mathbb{R}^3)$ and has no boundary distribution under the Laplacian. It is therefore an outgoing solution in the prescribed branch ball. Theorem 2.3 identifies it with $u_f(k)$. □

Now we study uniqueness of the inverse source problem at a single frequency. We begin by constructing examples showing that distinct sources may produce identical wave fields on the boundary. To do this, let $F_0 \in \mathcal{U}_k$ satisfy $T_k F_0 \neq 0$ and choose $0 \neq \psi \in C_c^\infty(D)$. For sufficiently small real t,

$$f_t = \Phi_k(F_0 + tL_k\psi) \in \mathcal{W}_k, \qquad g_{f_t}(k) = T_k F_0 \neq 0.$$

Indeed, $T_k L_k \psi = 0$, while $L_k\psi \neq 0$ by outgoing uniqueness. The injectivity of Φ_k makes the sources f_t distinct. Such families occur through every source in $\mathcal{W}_k$ having nonzero data.

To obtain the uniqueness, fix a nonzero real function $\chi \in C_c^\infty(D)$ such that $0 \le \chi \le 1$ and $\chi = 1$ in a neighborhood of $\operatorname{supp} q$, and define

$$\mathcal{R}_{k,\chi} = \{\chi v : v \in \mathcal{H}_k\} \subset C_c^\infty(D) \subset X, \tag{4.4}$$

where products are extended by zero. In the following theorem, we obtain a function space in which the source can be uniquely determined by data at a single frequency.

Theorem 4.5. *There exists $0 < \rho_{\mathrm{sf}} \le \rho_*$, independent of k, such that, for every $k > l$ and $0 < \rho \le \rho_{\mathrm{sf}}$, the map $f \mapsto g_f(k)$ is injective on $\mathcal{A}_\rho \cap \mathcal{R}_{k,\chi}$. In particular, every nonzero source in this class is radiating at k.*

Proof. Let $f_i = \chi v_i$, $v_i \in \mathcal{H}_k$, and suppose $g_{f_1}(k) = g_{f_2}(k)$. Put $v = v_1 - v_2$, $h = \chi v$, and

$$n = k^2 q\Big(|u_{f_1}(k)|^2 u_{f_1}(k) - |u_{f_2}(k)|^2 u_{f_2}(k)\Big).$$

The effective source difference $h - n$ lies in $\ker T_k = \mathcal{H}_k^\perp$. Pairing with v gives

$$J := \int_D \chi|v|^2 \,\mathrm{d}x = \int_D n\overline{v}\,\mathrm{d}x.$$

Since $\operatorname{supp} n \subset \operatorname{supp} q$ and $\chi = 1$ there, we have $\|v\|_{L^2(\operatorname{supp} q)} \le J^{1/2}$, and also $\|h\|_c \le c_+ J^{1/2}$ because $0 \le \chi \le 1$. Estimate (3.4) therefore yields

$$J \le \|n\|_{L^2(D)} J^{1/2} \le C\rho^2\langle k\rangle^{-3} \|h\|_c J^{1/2} \le Cc_+\rho^2\langle k\rangle^{-3} J.$$

Choose ρ_{sf} so that $Cc_+\rho_{\mathrm{sf}}^2 < 1$. Then we have $J = 0$ and $h = 0$. The proof is complete. □

Remark 4.6. The class in (4.4) contains nonzero sources. For $\omega \in \mathbb{S}^2$, the field

$$v_\omega = e^{ik\omega\cdot x} - \mathcal{S}_+(k)\Big[k^2(c^{-2} - 1)e^{ik\omega\cdot x}\Big]$$

satisfies $L_k v_\omega = 0$ in $\mathbb{R}^3$. It cannot vanish identically, since the second term is outgoing and the plane wave term is not. Unique continuation gives $\chi v_\omega \neq 0$. Moreover, we have $\int_D \chi|v_\omega|^2 \,\mathrm{d}x > 0$, and then Lemma 4.1 gives $T_k(\chi v_\omega) \neq 0$. Scaling v_ω by a sufficiently small nonzero constant places χv_ω in $\mathcal{A}_\rho$. Theorem 4.5 establishes uniqueness within this prescribed source class.

5. Numerical reconstruction

We derive a projected spectral iteration from Lemma 3.1 and then examine nonlinear correction and frequency enlargement in experiments.

5.1. **Computation of the outgoing field.** Since $c = 1$ and $q = f = 0$ outside B_R, the exterior field can be eliminated by the known outgoing Dirichlet-to-Neumann operator T_k^+ on Γ. Writing a trace in spherical harmonics as $g(R\omega) = \sum_{\ell,m} g_{\ell m} Y_{\ell m}(\omega)$ gives

$$(T_k^+ g)(R\omega) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} t_\ell(k) g_{\ell m} Y_{\ell m}(\omega), \qquad t_\ell(k) = k \frac{(h_\ell^{(1)})'(kR)}{h_\ell^{(1)}(kR)}. \tag{5.1}$$

The prime denotes differentiation of the spherical Hankel function with respect to its argument. This operator is computed from the homogeneous exterior. Define

$$a_k(u, v) = \int_{B_R} \left(\nabla u \cdot \nabla \overline{v} - k^2 c^{-2} u \overline{v} \right) \mathrm{d}x - \langle T_k^+ \operatorname{Tr} u, \operatorname{Tr} v \rangle_\Gamma.$$

The sign convention in (1.1) gives the weak equation

$$a_k(u, v) - k^2 \int_D q|u|^2 u \overline{v} \, \mathrm{d}x = - \int_D f \overline{v} \, \mathrm{d}x, \qquad v \in H^1(B_R).$$

Starting from $u^{(0)} = 0$, the Picard iteration is

$$a_k(u^{(r+1)}, v) = - \int_D f \overline{v} \, \mathrm{d}x + k^2 \int_D q|u^{(r)}|^2 u^{(r)} \overline{v} \, \mathrm{d}x. \tag{5.2}$$

For sources in the small branch, the continuous iteration is the resolvent contraction of Theorem 2.3. At an interior Dirichlet eigenfrequency the field problem still uses the outgoing condition (5.1). Imposing a zero Dirichlet condition instead would change the problem and introduce an interior resonance.

5.2. **Measured moments and a constrained source space.** The method assumes acquisition at the eigenfrequencies used below, as allowed in the discrete-data estimate (3.7). Measurements on a fixed band alone do not supply eigenfrequency traces outside that band. The present procedure does not replace those traces by analytic continuation. Choose N so that the first N eigenfunctions contain every eigenspace at the largest included eigenfrequency. Let λ_p be the distinct frequencies among $\kappa_1, \ldots, \kappa_N$, with multiplicities m_p. One measured trace at λ_p supplies all moments in its eigenspace. For data g^δ, form

$$b_j^\delta = - \int_\Gamma g^\delta(x, \kappa_j) \partial_\nu \overline{\phi_j}(x) \, \mathrm{d}S, \qquad 1 \le j \le N. \tag{5.3}$$

The normal derivatives in this expression belong to the known background eigenfunctions. No derivative of the measured trace is taken. The modes are obtained from the generalized Dirichlet eigenproblem $-\Delta \phi_j = \kappa_j^2 c^{-2} \phi_j$ and are normalized as in (3.1). Eigenfrequency, boundary flux, and quadrature errors should be assessed separately from measurement noise.

For a trial source f on the small branch, define its nonlinear correction

$$\mathfrak{e}_j(f) = \kappa_j^2 \int_D q|u_f(x, \kappa_j)|^2 u_f(x, \kappa_j) \overline{\phi_j}(x) \, \mathrm{d}x. \tag{5.4}$$

Green's identity gives $f_j = b_j(f) + \mathfrak{e}_j(f)$, with $b_j(f)$ defined by (5.3) using exact data. Each correction must be computed at its own eigenfrequency. A field solve is shared only among modes belonging to the same eigenspace.

Work in the source Hilbert space $\mathcal{H}_c = L^2(B_R, c^2\,\mathrm{d}x)$, with inner product $(f,g)_c = \int_{B_R} c^2 f\overline{g}\,\mathrm{d}x$. The functions $\psi_j = c^{-2}\phi_j$ are orthonormal in this space, and

$$P_N f = \sum_{j=1}^{N} f_j \psi_j, \qquad f_j = (f, \psi_j)_c.$$

The factor c^{-2} is essential in an inhomogeneous medium. A raw spectral sum need not have the prescribed support or whole-space H^4 regularity. Choose linearly independent functions $\zeta_1, \dots, \zeta_M \in C_c^\infty(D)$ and set

$$V_M = \operatorname{span}_{\mathbb{C}}\{\zeta_1, \dots, \zeta_M\}, \qquad \mathcal{C}_M = \{f \in V_M : \|f\|_{H^4(\mathbb{R}^3)} \le \rho\}. \tag{5.5}$$

This is a nonempty closed convex subset of $\mathcal{H}_c$. Denote its $\mathcal{H}_c$ metric projection by Π_M. The constraint ensures that all source iterates belong to $\mathcal{A}_\rho$. For implementation, fix the norm $\|f\|_{H^4}^2 = \sum_{|\alpha|\le 4} \|\partial^\alpha f\|_{L^2(\mathbb{R}^3)}^2$. Use this same convention for source normalization, the bound ρ, and the projection Gram matrix.

5.3. **Projected nonlinear spectral correction.** For a fixed frequency cutoff, the reconstruction is

$$f^{(r+1)} = \mathcal{T}_N^\delta(f^{(r)}) := \Pi_M \left[\sum_{j=1}^{N} \Big(b_j^\delta + \mathfrak{e}_j(f^{(r)}) \Big) \psi_j \right], \qquad f^{(0)} = 0. \tag{5.6}$$

Thus the first iterate is the constrained linear spectral reconstruction. Subsequent iterates correct its Kerr contribution using the full nonlinear forward solution. The following consequence of Lemma 3.1 provides a convergence guarantee for exact field solves, eigenpairs, integration, and projection.

Theorem 5.1. *Let $0 < \rho \le \rho_*$ and let $\theta < 1$ be the constant in Lemma 3.1. For each N and each moment vector $(b_j^\delta)_{j\le N}$, the map $\mathcal{T}_N^\delta$ is a contraction on $\mathcal{C}_M$ with constant at most θ. It has a unique fixed point $f_{N,M}^\delta$, and iteration (5.6) converges to it. If the true source $f^\dagger \in \mathcal{A}_\rho$, put*

$$\delta_b = \left(\sum_{j=1}^{N} |b_j^\delta - b_j(f^\dagger)|^2 \right)^{1/2}, \qquad d_M = \inf_{v\in\mathcal{C}_M} \|f^\dagger - v\|_c.$$

Then

$$\|f_{N,M}^\delta - f^\dagger\|_c \le \frac{\delta_b + \|(I-P_N)f^\dagger\|_c + d_M}{1-\theta} \le \frac{\delta_b + Q\langle \kappa_{N+1}\rangle^{-4} + d_M}{1-\theta}, \tag{5.7}$$

where $Q \ge \|f^\dagger\|_{\mathcal{X}}$. The contraction constant and the admissible source threshold are independent of N and M.

Proof. For $f, g \in \mathcal{C}_M$, Lemma 3.1 gives

$$\left(\sum_{j\le N} |\mathfrak{e}_j(f) - \mathfrak{e}_j(g)|^2 \right)^{1/2} \le \theta \|f - g\|_c.$$

Orthonormality of ψ_j and nonexpansiveness of Π_M prove the contraction assertion. The closed set $\mathcal{C}_M$ is complete, so Banach's fixed point theorem applies. For $f_* = f_{N,M}^\delta$, let z_* be the spectral sum inside the projection in (5.6). The exact spectral identity implies

$$\|z_* - f^\dagger\|_c \le \delta_b + \theta\|f_* - f^\dagger\|_c + \|(I-P_N)f^\dagger\|_c.$$

Since $\|\Pi_M f^\dagger - f^\dagger\|_c = d_M$, nonexpansiveness gives $\|f_* - f^\dagger\|_c \le \|z_* - f^\dagger\|_c + d_M$. Absorption and the spectral tail bound prove (5.7). □

The uniqueness in this theorem concerns the fixed point of the regularized reconstruction map. Finite frequency data can still admit several sources in the original inverse problem. The tail and d_M in (5.7) express the remaining ambiguity and approximation error. The algorithm requires no numerical analytic continuation of noisy data.

The projection in (5.6) reduces to a small convex problem. For $z \in \mathcal{H}_c$, set

$$G_{rs} = (\zeta_s, \zeta_r)_c, \qquad H_{rs} = (\zeta_s, \zeta_r)_{H^4(\mathbb{R}^3)}, \qquad d_r = (z, \zeta_r)_c.$$

Then $\Pi_M z = \sum_m a_m \zeta_m$, where

$$(G + \lambda H)a = d, \quad a^* H a \le \rho^2, \quad \lambda \ge 0, \quad \lambda(a^* H a - \rho^2) = 0.$$

If the unconstrained solution violates the bound, λ is determined by a scalar monotone search. This multiplier is fixed by the source constraint, rather than selected as a noise-dependent regularization parameter. This is the projection in the source $L^2(c^2\,\mathrm{d}x)$ norm, not in the Euclidean coefficient norm.

For an inexact implementation that preserves $\mathcal{C}_M$, write $\widetilde{f}^{(r+1)} = \mathcal{T}_N^\delta(\widetilde{f}^{(r)}) + \xi_r$, where $\|\xi_r\|_c \le \varepsilon_{\mathrm{alg}}$. Then

$$\|\widetilde{f}^{(r)} - f_{N,M}^\delta\|_c \le \theta^r \|\widetilde{f}^{(0)} - f_{N,M}^\delta\|_c + \frac{\varepsilon_{\mathrm{alg}}}{1-\theta}.$$

Here $\varepsilon_{\mathrm{alg}}$ includes forward, spectral, integration, and projection errors measured against the exact map. It must be controlled by independent refinement, not inferred from a small iteration increment alone. For the exact iteration, $\|f^{(r)} - f_{N,M}^\delta\|_c \le \theta(1-\theta)^{-1}\|f^{(r)} - f^{(r-1)}\|_c$ for $r \ge 1$.

5.4. **Experimental model, sources, and fixed reconstruction spaces.** We test (1.1) with a nonzero Kerr coefficient, $R = 1$, $D = B_{0.9}$, and

$$m(r) = c^{-2}(r) = 1 + 0.2 b_{0.7}(r), \qquad q(r) = 0.5 b_{0.65}(r).$$

$$b_s(r) = \begin{cases} \exp[-r^2/(s^2 - r^2)], & 0 \le r < s, \\ 0, & r \ge s. \end{cases}$$

For translated bumps we use $b_s(x) = b_s(|x|)$. The two complex sources are

$$\begin{aligned} F_A(x) &= b_{0.28}(x - (-0.28, 0, 0)) + (0.7 + 0.4i) b_{0.24}(x - (0.25, 0.12, 0)), \\ f_A &= 0.1 F_A / \|F_A\|_{H^4}, \\ F_O(r, \omega) &= b_{0.75}(r)\Big[(0.7\cos(28r) + \cos(95r)) Y_{00}(\omega) \\ &\quad + (0.7 + 0.25i) r \cos(145r) Y_{11}(\omega) + (0.5 - 0.4i) r^2 \cos(60r) Y_{2,-1}(\omega) \\ &\quad + 0.3 r^2 \cos(175r) Y_{22}(\omega)\Big], \\ f_O &= F_O / \|F_O\|_c. \end{aligned}$$

Here $Y_{\ell m}$ are orthonormal complex spherical harmonics. The H^4 norm is the convention specified after (5.5). Both sources are smooth and supported in $B_{0.75}$. We retain these original amplitudes and also consider the same shapes scaled to $\|f\|_c = 5$. Thus the four cases are f_A, f_O, $5f_A/\|f_A\|_c$, and $5f_O$, where $\|f_A\|_c \simeq 3.90357 \times 10^{-8}$ and $\|f_O\|_c = 1$. The amplified cases make the nonlinear response visible. Their amplitudes were selected from forward convergence and response diagnostics.

The fixed source spaces use bump-weighted Chebyshev radial functions, with additional cosine functions at low angular degrees, followed by orthonormalization with relative singular-value threshold 10^{-10}. For the double bump we retain $\ell \le 20$, giving $M = 4731$. For the oscillator we use its declared family prior $\ell \le 2$, giving $M = 459$. Each source space is fixed across all amplitudes, noise levels, and available frequency cutoffs. The oscillator's

angular support is prior information, not identified blindly. The radial dictionary is also informed by the benchmark: the carriers 60, 95, 145, and 175 lie on its grid, whereas 28 does not. These tests therefore do not represent blind recovery of arbitrary H^4 sources. The double-bump relative best-approximation error is 3.450483%. The component at $\ell > 20$ alone has relative norm 1.138505%. Both contributions are included in the reported global errors.

5.5. **Forward data, noise, and nonlinear reconstruction.** The available cutoffs are $K = 10, 20, 50, 100, 150, 200$. The acquisition uses the angular multiplets

$$\mathcal{J}_{K,20} = \{(\ell, n, m) : \kappa_{\ell n} \le K,\ 0 \le \ell \le 20,\ |m| \le \ell\}.$$

At $K = 200$ this gives 1276 distinct frequencies and 26026 spectral coordinates. These counts differ from the source dimension M and do not form the complete three-dimensional Dirichlet spectrum below K.

Forward fields use fourth-order radial Numerov differences and the outgoing spherical Hankel ratio. The synthetic trace combines a fine linear baseline with a nonlinear defect:

$$g_q^{\mathrm{data}} = g_0^{8192,80} + \left(g_q^{2048,24} - g_0^{2048,24}\right).$$

The superscripts give radial intervals and field angular degree. The bracket is obtained by nonzero-q forward solves. Inverse fields use 1024 radial intervals and angular degree 20, so the forward and inverse discretizations differ.

The trace is sampled at 96 Gauss nodes in $\cos\vartheta$ and 192 azimuthal nodes, giving 18432 sensors per frequency. For $\alpha \in \{0.01, 0.02, 0.05\}$, circular complex Gaussian draws ξ_p are normalized by

$$\eta_p^\alpha = \alpha \frac{\|g_q^{\mathrm{data}}(k_p)\|_W}{\|\xi_p\|_W}\xi_p, \qquad g_p^\delta = g_q^{\mathrm{data}}(k_p) + \eta_p^\alpha, \qquad \|z\|_W^2 = \sum_s w_s |z_s|^2.$$

The noise norm is set using the complete boundary trace before angular projection. One seed, 72451, is paired across sources, amplitudes, noise levels, and frequency prefixes. All errors are single realizations.

We distinguish the available cutoff K from the used cutoff $\kappa \le K$. Among the candidates not exceeding K, a Lepskii comparison selects the smallest κ such that, for every larger available candidate t,

$$\|a_t^0 - a_\kappa^0\|_2 \le 2(\sigma_t + \sigma_\kappa) + 10^{-5} \max_{s \le K} \|a_s^0\|_2.$$

Here a_s^0 is the noisy linear coefficient vector and σ_s is its RMS noise budget, calibrated from the sensor noise and measurement responses. This is a calibrated RMS heuristic. It selects $\kappa = \min(K, 100)$ for both double-bump amplitudes at all three noise levels, and $\kappa = K$ for both oscillatory amplitudes.

Each reconstruction update solves the nonzero-q field for the current trial source, evaluates (5.4), and projects the updated spectral sum onto its fixed V_M. True source moments are not supplied to this iteration. The implementation omits the H^4 ball projection in (5.6). It is an unconstrained variant of the proved algorithm. The complete selected-cutoff discrete map is checked with relative residual at most 2×10^{-7}.

5.6. **Reconstruction results and error reduction.** We compare four amplitude settings, six available cutoffs, and three noise levels. Errors are measured by $\|\widehat{f} - f\|_c / \|f\|_c$ over the full three-dimensional source, including angular omissions and source-space approximation. Table 1 reports errors at the largest available cutoff.

The double-bump error approaches its fixed-space approximation floor 3.450483% at $\kappa = 100$, whereas the oscillator continues to recover fine structure through $\kappa = 200$. At

TABLE 1. Relative source errors in percent at available $K = 200$. The three middle error columns are Kerr-corrected. The last column gives the linear initial error at 2% noise.

			Kerr-corrected			Linear initial
Case	M	κ	1%	2%	5%	2%
Original f_A	4731	100	3.4516	3.4527	3.4607	3.4527
Original f_O	459	200	0.1648	0.1689	0.1951	0.4634
Amplified f_A	4731	100	3.4515	3.4527	3.4604	6.1110
Amplified f_O	459	200	0.1648	0.1690	0.1959	11.0048

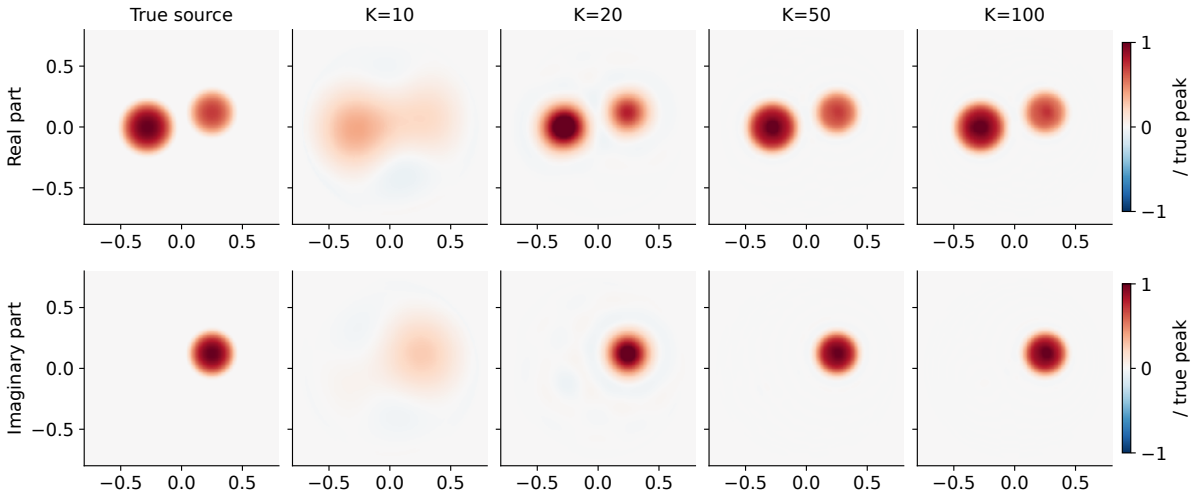


FIGURE 2. Amplified double-bump source at representative cutoffs. For available $K \geq 100$, the same reconstruction with $\kappa = 100$ is used.

$K = 200$ and 2% noise, the Kerr correction reduces the amplified oscillator error from 11.0048% to 0.1690%. The original double bump has negligible nonlinear response at the displayed precision.

Figures 2 and 3 show real and imaginary $z = 0$ sections at 2% noise, normalized by the corresponding true-component peak. Figure 4 compares linear and Kerr-corrected errors using the same source space, noisy data, and selected cutoff. The global error includes every source component, including $Y_{2,-1}$, which vanishes on the displayed equatorial plane.

5.7. **Verification and scope.** Independent checks compare the data, source errors, and reconstruction maps. Radial and angular refinement at representative frequencies assesses the trial-source correction moments. These checks include less-resolved high-angular forward-defect components and their effects on retained moments. No uniform continuous frequency error bound is inferred.

Sensor projection and the declared source prior reduce effective moment noise relative to physical sensor noise. Subpercent errors at 5% therefore concern this noise model, rather than arbitrary perturbations. The computations illustrate Kerr correction and frequency enlargement without certifying the continuous smallness threshold, stability constants, a complete three-dimensional K^{-4} rate, or the fixed-band double-logarithmic estimate.

Additional implementation details, reconstruction results, and verification records are provided in the supplementary material.

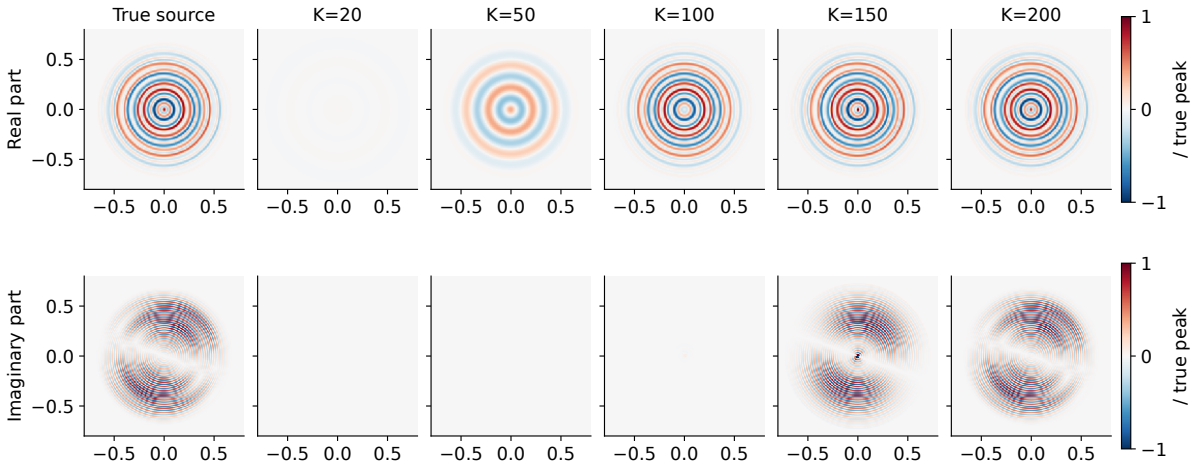


FIGURE 3. Amplified oscillatory source with $\kappa = K$. The columns show the recovery of fine structure as frequency increases.

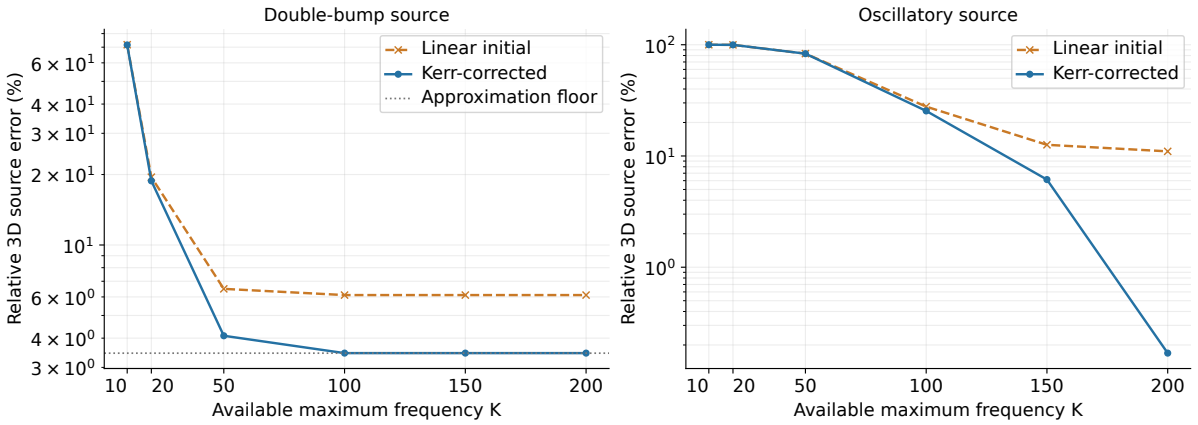


FIGURE 4. Linear initial and Kerr-corrected errors for the amplified sources at 2% physical noise.

6. CONCLUSION

We have established multifrequency uniqueness and conditional stability for small sources on the outgoing branch in a nontrapping inhomogeneous Kerr medium. Full-spectrum control of the nonlinear remainder yields finite-eigenfrequency estimates with decreasing spectral tails and a Lipschitz estimate for the complete moment sequence. Fixed-band data give uniqueness and double-logarithmic stability. At a fixed frequency, effective-source coordinates describe the data fibers, and uniqueness holds on a prescribed source class. The same remainder estimate gives a uniformly contractive projected reconstruction, with an exact-operator error bound separating moment noise, spectral truncation, and source-space approximation. Nonzero-Kerr experiments show substantial improvements from nonlinear correction for amplified sources. The double-bump reconstruction approaches its source-space approximation floor, while the oscillatory source gains resolution as the used frequency range increases. Reducing the source regularity required by the analysis remains an open direction.

Appendix A. Two useful lemmas

The first lemma quantifies analytic continuation from a fixed frequency interval. The second one collects the spectral estimates used in Section 3.

Lemma A.1. *Let Y be a complex Hilbert space and let $F : \Omega_d \to Y$ be holomorphic, where*

$$\Omega_d = \{z \in \mathbb{C} : \operatorname{Re} z > l,\ |\operatorname{Im} z| < d\}.$$

Suppose that $I = (a, b)$, $0 < l < \min\{a, \kappa_1\}$, and

$$\sup_{z\in\Omega_d} \|F(z)\|_Y \le M, \qquad \sup_{k\in I} \|F(k)\|_Y \le \epsilon, \qquad 0 < \epsilon < M.$$

There exist constants $A_0, B_0 > 0$, depending only on l, d, I, and κ_1, such that

$$\|F(k)\|_Y \le M(\epsilon/M)^{A_0 e^{-B_0 K}}, \qquad \kappa_1 \le k \le K. \tag{A.1}$$

Proof. Fix a nondegenerate closed interval $J \subset I$, and let ω be the harmonic measure of J in $\Omega_d \setminus J$. It can be constructed by increasing exhaustion with bounded slit rectangles, assigning the boundary values one on J and zero on the outer boundary. The maximum principle gives $0 < \omega \le 1$ in the slit domain. Moreover, ω extends continuously to J with value one. The endpoints of a line segment are regular boundary points, as follows from a square-root barrier.

For $\ell \in Y^*$ with $\|\ell\| \le 1$, the two-constants theorem on each bounded slit rectangle, followed by exhaustion, gives

$$|\ell(F(z))| \le M(\epsilon/M)^{\omega(z)}, \qquad z \in \Omega_d \setminus J. \tag{A.2}$$

Indeed, this is obtained by applying the maximum principle to the subharmonic function $\log(|\ell(F)|/M)$, whose boundary bounds are $\log(\epsilon/M)$ on J and zero on the outer boundary.

We give a lower bound for ω on the positive real axis. Choose $x_0 > \max\{b, \kappa_1\}$ and $r > 0$ so small that every disk of radius $2r$ centered at a real point $x \ge x_0$ lies in $\Omega_d \setminus J$. Harnack's inequality on these disks implies

$$\omega(x + t) \ge \tfrac{1}{3}\omega(x), \qquad x \ge x_0, \quad 0 \le t \le r.$$

A chain of at most $1 + (k - x_0)/r$ disks therefore yields

$$\omega(k) \ge \tfrac{1}{3}\omega(x_0) \exp\Big(-r^{-1}(\log 3)(k - x_0)\Big), \qquad k \ge x_0.$$

On the compact segment $[\kappa_1, x_0]$, extend ω by one on J. Positivity and continuity give a positive minimum there. Consequently, after decreasing A_0 if necessary, we have

$$\omega(k) \ge A_0 e^{-B_0 K}, \qquad \kappa_1 \le k \le K, \qquad A_0 e^{-B_0 K} \le 1.$$

Combining this with (A.2) and taking the supremum over ℓ proves (A.1) off J. On J, the assumed measurement bound gives the same conclusion. □

Lemma A.2. *Let P_D be the Dirichlet realization of $-c^2\Delta$ on B_R, and let $\{\phi_j\}$ be an orthonormal eigenbasis in $L^2(B_R, c^{-2}\,\mathrm{d}x)$, with eigenvalues κ_j^2 ordered with multiplicity. There are constants $C, c_1, c_2 > 0$ such that*

$$\|\partial_\nu \phi_j\|_{L^2(\Gamma)} \le C\kappa_j, \qquad c_1 j^{1/3} \le \kappa_j \le c_2 j^{1/3}, \qquad j \ge 1. \tag{A.3}$$

Proof. Write $m = c^{-2}$ and $\lambda = \kappa_j^2$. The eigenfunction equation is $-\Delta\phi_j = \lambda m \phi_j$, and its energy identity is

$$\int_{B_R} |\nabla\phi_j|^2\,\mathrm{d}x = \lambda \int_{B_R} m|\phi_j|^2\,\mathrm{d}x = \lambda.$$

Multiplication by $x \cdot \nabla\overline{\phi_j}$, integration by parts, and taking twice the real part give the Rellich identity

$$\int_\Gamma (x \cdot \nu)|\partial_\nu \phi_j|^2 \, \mathrm{d}S = \lambda \int_{B_R} (2m + x \cdot \nabla m)|\phi_j|^2 \, \mathrm{d}x.$$

Here the tangential derivative vanishes on Γ. Since $x \cdot \nu = R$ on Γ, m is smooth and bounded below, and $\int m|\phi_j|^2 = 1$, the right-hand side is bounded above by $C\lambda$. This proves the boundary estimate $\|\partial_\nu \phi_j\|_{L^2(\Gamma)} \le C\kappa_j$.

Let μ_j be the eigenvalues of the ordinary Dirichlet Laplacian on B_R. The Rayleigh quotient for P_D is

$$\frac{\int_{B_R} |\nabla v|^2 \, \mathrm{d}x}{\int_{B_R} c^{-2}|v|^2 \, \mathrm{d}x}, \qquad 0 \neq v \in H_0^1(B_R).$$

The bounds $c_- \le c \le c_+$ and the min–max principle yield $c_-^2 \mu_j \le \kappa_j^2 \le c_+^2 \mu_j$. The Dirichlet Weyl law in dimension three gives $\mu_j \asymp j^{2/3}$, and hence the remaining estimates in (A.3). □

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Conflict of Interest Statement

The authors declare that they have no conflict of interest.

Data Availability Statement

The numerical experiments use synthetic data generated using the models and parameters described in this article. The data supporting the numerical results are available from the corresponding author upon reasonable request.

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